



6.7848	98.311	16.0213	722.504
4.6323	65.28	18.0289	757.512
		21.0388	796.328
		24.0253	825.825

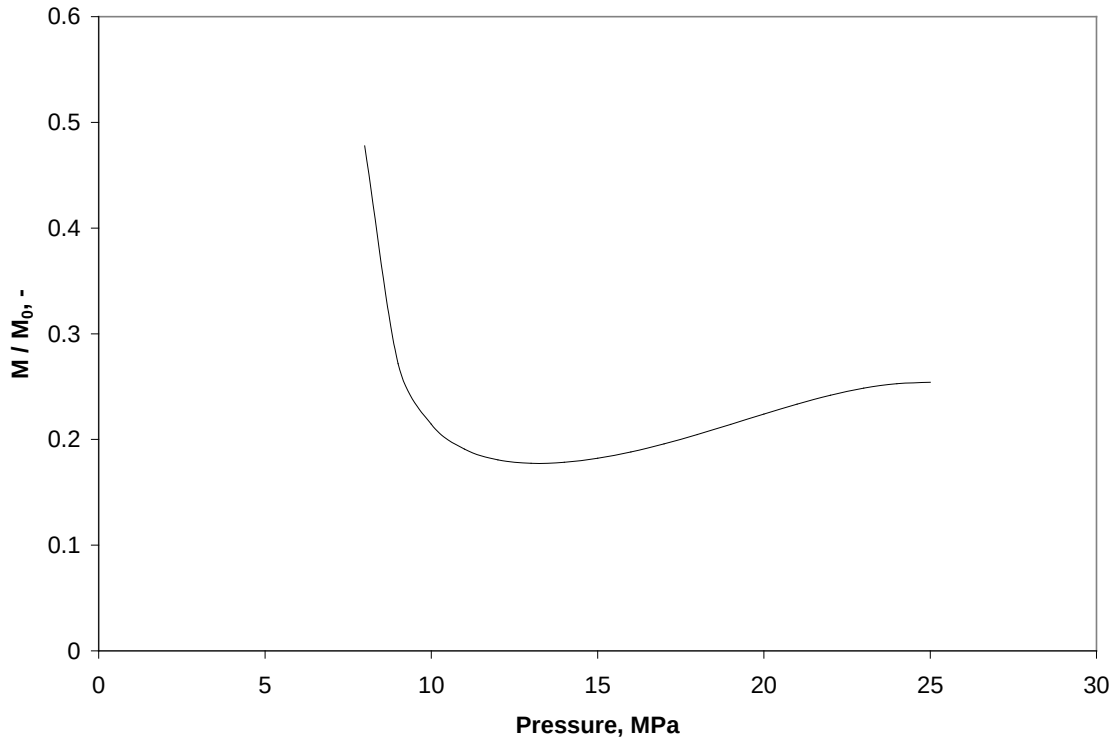


Figure 2.5. Verification of the existence of a minimum in the normalized density for impure CO₂, hence in storage capacity, using literature data given in Table 2.1.

2.2.4 A Maximum of Storage Capacity for CO₂ with SO₂

For impurities which have higher critical temperature than that of CO₂, no storage capacity minimum would occur in the corresponding CO₂ mixture. Rather, a maximum can appear based on an analysis similar to that for the non-condensable impurities. This has been verified with calculated results for the CO₂ mixture with 2.9 vol % SO₂ (the Cansolv[®] absorption system case presented in the previous IEA GHG report PH4/32). As shown in Figure 2.6, a maximum is seen at about 11 MPa, where the storage capacity is increased by over 5%. It should be noted again that the storage capacity is based on pure CO₂ (see discussion for Equation 2.5). It is, therefore, interesting to see that in this peak region, the CO₂-based capacity is above that of pure CO₂. In other words, SO₂ does not take any but creates space for CO₂. This can be rationalized from the consideration that SO₂ decreases average distance between the molecules of the mixture – an opposite effect of that of the non-condensable gases. This can be taken as one positive effect of the presence of SO₂ on CO₂ storage. However, SO₂ causes more concern with regard to its chemical effects on formation rocks. This will be discussed later.